



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2025

Round 1: Online Knowledge Challenge

Junior Division (ages 12-14)

Mark scheme and worked solutions

Total: 50 marks. Accept answers within rounding and correct alternative methods. Apply error carried forward (ECF): if a wrong value from an earlier part is used correctly in a later part, award the method marks.

Part A: Answer key

Question	Answer	Correct option
A1	B	94.9 kWh
A2	D	Tidal
A3	C	6
A4	C	10 kJ
A5	D	About 95%
A6	C	Water vapour
A7	C	Polypropylene, a bottle cap (0.90 g cm^{-3})
A8	C	435 kg
A9	B	Meadow X has higher biodiversity because the species are more evenly spread
A10	B	Most nutrients are stored in the living plants, and heavy rain washes the rest out of the thin soil

Part A: Worked solutions

A1. Answer B

Power saved = $60 - 8 = 52 \text{ W}$. Energy = $52 \text{ W} \times 5 \text{ h} \times 365 = 94\,900 \text{ Wh} = \mathbf{94.9 \text{ kWh}}$.
Option C is the total energy used by the old bulb, not the saving.

A2. Answer D

The Sun heats the air unevenly (wind), evaporates water that later flows downhill (hydro) and powers photosynthesis (wood). **Tides** are caused mainly by the Moon's gravity acting on a rotating Earth.

A3. Answer C

The balanced equation shows **6** O_2 molecules for every one glucose molecule.

A4. Answer C

$100\,000 \rightarrow 10\,000$ (grasshopper) $\rightarrow 1000$ (frog) $\rightarrow 100$ (snake) $\rightarrow \mathbf{10 \text{ kJ}}$ (hawk). There are four transfers, not five.

A5. Answer D

Extracting aluminium from its ore needs huge amounts of electricity for electrolysis. Melting down scrap needs only about 5% of that, saving about **95%**.

A6. Answer C

Water vapour is the most abundant greenhouse gas. Its amount is controlled by temperature, so it amplifies warming caused by CO_2 rather than starting it.

A7. Answer C

Only materials less dense than seawater float: **polypropylene** ($0.90 < 1.025$). This is why bottle caps are often found on beaches while the bottles sink, unless air is trapped inside.

A8. Answer C

Return distance = $2 \times 1450 = 2900$ km. $2900 \times 0.15 = \mathbf{435 \text{ kg}}$. Option B is a one-way trip.

A9. Answer B

Biodiversity depends on richness (number of species) and evenness (how equally they are represented). Richness is equal, but **X is far more even**, so it is more diverse.

A10. Answer B

In rainforests, nutrients are recycled quickly and held mainly in the vegetation. Once the trees are removed, heavy rainfall **leaches** the few remaining nutrients from the thin topsoil.

Part B: Worked solutions and mark scheme

B1. Computers left on (8 marks)

(a) [2 marks] $12 \times 60 \text{ W} \times 18 \text{ h} = 12\,960 \text{ Wh} = \mathbf{12.96 \text{ kWh}}$.

Marking: 1 mark for the method; 1 mark for converting to kWh.

(b) [2 marks] $12.96 \times 190 = \mathbf{2462 \text{ kWh}}$ (2462.4 kWh).

Marking: Allow error carried forward.

(c) [2 marks] $2462.4 \times \text{€}0.20 = \mathbf{\text{€}492}$ (€492.48).

Marking: Allow error carried forward.

(d) [2 marks] $2462.4 \times 0.25 = \mathbf{616 \text{ kg}}$ (615.6 kg) of CO₂.

Marking: Allow error carried forward.

B2. From food waste to compost (8 marks)

(a) [1 mark] $15 \times 190 = \mathbf{2850 \text{ kg}}$.

Marking: 1 mark.

(b) [2 marks] The mass falls by 60%, so 40% remains: $2850 \times 0.40 = \mathbf{1140 \text{ kg}}$.

Marking: 1 mark for using 40%; 1 mark for the answer.

(c) [3 marks] Methane = $2850 \times 0.05 = 142.5 \text{ kg}$. CO₂-equivalent = $142.5 \times 28 = 3990 \text{ kg} \approx \mathbf{4.0 \text{ t}}$.

Marking: 1 mark for methane; 1 mark for multiplying by 28; 1 mark for the answer.

(d) [2 marks] Methane is made by microbes that live **without oxygen** (anaerobic decomposition), as in buried landfill waste. A compost heap that is turned regularly contains oxygen, so microbes break the waste down aerobically into CO₂ and water instead.

Marking: 1 mark for anaerobic conditions in landfill; 1 mark for oxygen/aerobic in the compost heap.

B3. Cycling to school (7 marks)

(a) [2 marks] $8.0 \times 190 = 1520 \text{ km}$. $1520 \times 0.120 \text{ kg} = \mathbf{182.4 \text{ kg}}$.

Marking: 1 mark for distance; 1 mark for the answer in kg.

(b) [2 marks] $30 \times 182.4 = 5472 \text{ kg} \approx \mathbf{5.5 \text{ t}}$.

Marking: Allow error carried forward.

(c) [3 marks] Car: $8.0/30 \text{ h} = 16 \text{ min}$. Bike: $8.0/15 \text{ h} = 32 \text{ min}$. Extra time = **16 minutes per day**.

Marking: 1 mark for each time; 1 mark for the difference.

B4. Reading climate data (7 marks)

(a) [3 marks] 1960–1980: $22/20 = \mathbf{1.10}$; 1980–2000: $31/20 = \mathbf{1.55}$; 2000–2020: $44/20 = \mathbf{2.20 \text{ ppm per year}}$.

Marking: 1 mark each.

(b) [2 marks] The concentration is not just rising: the **rate of increase is itself increasing** (it doubled over the 60 years). The rise is accelerating.

Marking: 1 mark for increasing concentration; 1 mark for the increasing rate.

(c) [2 marks] $414 + 10 \times 2.20 = \mathbf{436 \text{ ppm.}}$

Marking: Allow error carried forward.